

(2003) computes human capital returns for various professions and genders. He finds white males with one to five years of experience and a college degree have annual human capital returns of 14.2% versus 10.5% for those with some college but no degree. While college graduates have higher human capital returns, they are more volatile: the standard deviation of human capital returns for white male college graduates is 11.3% versus 4.7% for those with post-college education. Thus, the raw Sharpe ratio (which I define as the expected return [not excess] divided by volatility; see chapter 2) of white college males with one to five years of work experience is $14.2\%/11.3\% = 1.3$, which is lower than the 2.2 raw Sharpe ratio of the same demographic group with post-college experience. In comparison, the S&P 500 has a raw Sharpe ratio of around 0.6. Education is a very good investment, on average.

In addition to the risk of your human capital varying over time, there is also large disparity in human capital across professions. Christiansen, Joensen, and Nielsen (2007) report the mean human capital return for doctors is over 25% with a standard deviation of 18%. Human capital returns for masters graduates in economics have a mean of 12% and a standard deviation of 16%. Christiansen, Joensen, and Nielsen find that some professions have significantly negative returns, like individuals with master degrees in education or music.

Wages can fall a lot during recessions.¹⁴ Guvenen, Song, and Ozkan (2012) study the income dynamics of millions of individuals using Social Security Administration records. They find that during recessions, big upward movements of wages become very unlikely while severe drops in income become very likely. The middle part of the distribution tends to be fairly constant compared to the movement tails. The falls in income for the most wealthy—the top 1% (probably you, dear reader)—have the highest labor income risk. During the Great Recession in 2008–2009, the top 1% experienced enormous income losses, which were 21% worse, on average, than the losses of those in the ninetieth percentile.

Computing human capital returns using the growth in wages, however, actually understates the riskiness of human capital. Wages for individuals are like dividends for stocks. Stock prices can soar and plummet while their dividends remain fairly constant. Measuring stock returns by changes in dividends would produce very poor approximations to actual stock returns. In valuing stocks, we need to use *time-varying discount rates*. Likewise, the discount rates on human capital are not constant.¹⁵ Human capital can be very risky.

¹⁴ The major risk is that when you lose your job in a recession, your wage *permanently* falls. See Meghir and Pistaferri (2011) for a summary of permanent and temporary components in labor income.

¹⁵ As shown by Heaton and Lucas (2000).